

PROJECT LOOP

AI Customer Feedback Intelligence Platform - Technical Documentation

Author: Rakesh R

Team No: 16

Google Gemini 3.6 Flash

Neon PostgreSQL

Next.js 16

Project Title	Project LOOP - AI Customer Feedback Intelligence Platform
Author / Student Lead	Rakesh R
Team Number	Team No. 16
Version & Status	v1.0.0 (Production Release & Verified)
GitHub Repository	https://github.com/Rakesh-0804/loop.git
Live Deployment URL	https://loop-kappa-six.vercel.app

1. Executive Summary & Problem Statement

Project LOOP is an enterprise-grade AI customer feedback intelligence system designed to aggregate, analyze, and synthesize raw multi-channel customer feedback into real-time actionable product strategy insights.

Core Objective: Replace fragmented spreadsheet feedback reviews with instant, decisive AI sentiment classification (Positive, Neutral, Negative), continuous satisfaction scoring (0.05 to 0.98), automated theme clustering, and C-suite executive report generation.

2. System Architecture & Tech Stack

The platform is architected on a modern serverless stack providing sub-second query performance and seamless scalability:

- Frontend Framework:** Next.js 16 (App Router, Turbopack, React 19, Tailwind CSS).
- AI Engine:** Google Gemini 3.6 Flash (via @google/genai SDK v2.17+).
- Database:** Neon PostgreSQL (AWS Cloud Pooler Connection).
- ORM & Adapter:** Prisma 7.9.0 with @prisma/adapters driver adapter.
- Microservices API:** Python Flask 3.0+ backend connected via SQLAlchemy.

- **Authentication & Security:** NextAuth v5 (JWT Session, bcryptjs hashing, RBAC middleware).
- **UI & Aesthetics:** Glassmorphic design with unique per-page background particle visual canvas rendering (Dashboard, Inbox, Analytics, Themes, Reports, Settings).

3. Database Entity-Relationship (ER) Schema

The relational database schema is deployed on Neon PostgreSQL with enforced cascading references:

```
CREATE TYPE "Role" AS ENUM ('ADMIN', 'ANALYST', 'VIEWER');
CREATE TYPE "Sentiment" AS ENUM ('POS', 'NEU', 'NEG');
CREATE TYPE "FeedbackStatus" AS ENUM ('NEW', 'REVIEWED', 'ACTIONED');

CREATE TABLE "Workspace" (
  "id" TEXT PRIMARY KEY,
  "name" TEXT NOT NULL,
  "createdAt" TIMESTAMP(3) DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE "User" (
  "id" TEXT PRIMARY KEY,
  "name" TEXT NOT NULL,
  "email" TEXT UNIQUE NOT NULL,
  "passwordHash" TEXT NOT NULL,
  "role" "Role" DEFAULT 'ADMIN',
  "workspaceId" TEXT NOT NULL REFERENCES "Workspace"("id") ON DELETE CASCADE
);

CREATE TABLE "Feedback" (
  "id" TEXT PRIMARY KEY,
  "content" TEXT NOT NULL,
  "channel" TEXT NOT NULL,
  "sourceRef" TEXT,
  "customerLabel" TEXT,
  "sentiment" "Sentiment",
  "sentimentScore" DOUBLE PRECISION,
  "status" "FeedbackStatus" DEFAULT 'NEW',
  "createdAt" TIMESTAMP(3) DEFAULT CURRENT_TIMESTAMP,
  "workspaceId" TEXT NOT NULL REFERENCES "Workspace"("id") ON DELETE CASCADE
);

CREATE TABLE "Report" (
  "id" TEXT PRIMARY KEY,
  "title" TEXT NOT NULL,
  "periodStart" TIMESTAMP(3) NOT NULL,
  "periodEnd" TIMESTAMP(3) NOT NULL,
  "contentJson" TEXT NOT NULL,
  "createdAt" TIMESTAMP(3) DEFAULT CURRENT_TIMESTAMP,
  "workspaceId" TEXT NOT NULL REFERENCES "Workspace"("id") ON DELETE CASCADE,
  "generatedBy" TEXT NOT NULL
);
```

4. Key Core Features & Modules

4.1 Multi-Channel Feedback & Bulk CSV Ingestion

Supports feedback ingestion across 5 channels (Support Tickets, App Store Reviews, NPS Surveys, Sales Call Notes, Community Posts) plus bulk CSV file upload with automated batch Gemini AI scoring.

4.2 Decisive AI Sentiment & Theme Analysis

Decisive Gemini 3.6 Flash engine evaluates text without defaulting to neutral, assigning positive rating scores (0.70-0.98), negative rating scores (0.05-0.35), and matching theme categories.

4.3 Executive Synthesis & Multi-Format Exports

Generates detailed C-suite reports featuring CSAT satisfaction scores, Net Promoter Index, channel breakdown bars, churn risk callouts, customer quote excerpts, and strategic roadmaps. Supports PDF, CSV, and JSON exports.

5. Test System Credentials

Role	Email Address	Password	Permissions & Scope
ADMIN	admin@projectloop.ai	password123	Full Administrative, User Management & System Rights
ANALYST	analyst@projectloop.ai	password123	Read/Write Feedback, Bulk Ingestion & Executive Reports
VIEWER	viewer@projectloop.ai	password123	Read-Only Dashboard, Analytics & Report Access

6. API Endpoint Reference

Method	Endpoint Path	Service	Description
GET / POST	/api/feedback	Next.js API	Fetch or create feedback items with AI scoring
POST	/api/feedback/bulk-upload	Next.js API	Bulk CSV upload & batch Gemini AI sentiment classification
GET / POST	/api/reports	Next.js API	Fetch or synthesize executive intelligence reports
GET	/api/analysis	Next.js API	Compute multi-axis CSAT, NPS, and churn risk metrics
POST	/submit-feedback	Python Flask	Analyze single feedback via Flask backend
POST	/bulk-upload-csv	Python Flask	Process CSV item array via Flask AI engine